Exploring the Relationship Between Sex and Self-Reported Emotional Suppression

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Introduction

Emotional regulation is crucial to maintaining a healthy mind and body, yet society has created high expectations for each sex that make it difficult to practice healthy coping mechanisms. This can consequently lead to unhealthy coping alternatives, such as emotional suppression. Emotional suppression is the act of pushing away or ignoring uncomfortable thoughts and feelings, preventing an individual from consciously experiencing their emotions. Although emotional suppression does not conform to a specific gender, recent literature has found that emotional suppression is more relevant in one sex than the other.

A core aspect of masculinity revolves around the concept that men are discouraged from expressing their emotions, in fear of being perceived as weak and less masculine. As a result, men may be more skilled at suppressing their emotions as a coping mechanism (Martin, 2024) compared to women. Along with social norms placing a heavy burden on men to act a certain way, an individual's culture and family are another factor that leads to the act of emotional suppression. One study hypothesized that families may hold certain expectations on daughters and sons regarding their sex. Where daughters are expected to express their emotions more freely but are perceived as unstable compared to sons who are expected to be emotionally stable, tough, and not that expressive (Caui, et al., 2016). "Therefore, males may unconsciously follow the norms of masculinity to suppress their unpleasant emotions in daily life, consequently leading to a greater skill at regulating unpleasant emotion by suppressing emotional expression" (Caui, et al., 2016). This supports the idea that men are more inclined to suppress their emotions as they have been taught at a young age to avoid showing vulnerability and conform to social expectations.

Various studies have shown that “infant boys have a higher activity and arousal levels, and lower language ability and inhibitory control than girls” (Chaplin, 2015). Demonstrating that their biology already places these young boys at a great disadvantage, as their low abilities such as language and inhibitory control make it difficult to regulate and communicate their emotions efficiently. Furthermore, these clear biological differences can also result in different upbringings on how these sexes may address their emotions. Where boys are exposed to an environment that encourages them to limit their emotional expression with the means to down-regulate their high arousal and activity levels. While girls with lower arousal levels and higher vocabulary are encouraged to express their emotional state (Chaplin, 2015). Since females are encouraged to express their emotions at such an early age, they are less likely to suppress their emotions and may lean towards other coping strategies to regulate their emotions.

The validity of a possible relationship between emotional suppression and sex is supported by the following information that suggests biological and social factors have a role behind the sexes' different approaches to emotional regulation. This paper will further explore the relationship between sex and emotional suppression to support the hypothesis that males are more inclined to suppress their emotions compared to females.

Method

Participants

Participant data was collected through the study, “Midlife in the United States (MIDUS 3): Neuroscience Project, 2017-2022” led by principal investigators Carol D. Ryff and Richard J. Davidson at the University of Wisconsin-Madison. From the 170 total participants, our sample size will primarily focus on the 169 participants who answered the “Emotion Regulation

Questionnaire (ERQ)". Participants's age ranged from 43 to 87 (M = 59.82). This paper will analyze the correlation between the mean ERQ score for "Suppression" and the participants' sex group: male and female.

Participation in MIDUS 3 was voluntary. For those who participated, they were paid $12 prior to the study and an additional $50 after completing the whole series of surveys.

Procedure

The high success rate of the "Midlife Development in the U.S. (MIDUS)" study led to its third extension in 2013 known as MIDUS 3. A longitudinal study aimed to examine America's health and well-being during changing times through baseline assessments such as phone interviews and self-administered questionnaires (Ryff et al., 2013, 2017). Among the various self-administered questionnaires given is an "Emotion Regulation Questionnaire (ERQ)" which asks participants about their emotional life and how they control their emotions. Using a 7-point Likert scale with 1 being "Strongly Disagree" and 7 as "Strongly Agree", participants then selected the score that best represented them. In addition to the ERQ, participants were also asked about their demographics and background such as gender, age, education, and more.

Data Analysis

An independent sample *t*-test will be conducted on the two variables: participant's suppression score and their sex. The suppression mean score is a continuous variable that ranges from 1 to 6.50, where a mean of 1 represents the number of participants who selected a low score and a mean of 6.5 represents the number of participants who selected a high score. Sex is a binary variable with two independent and non-overlapping variables: male and female.

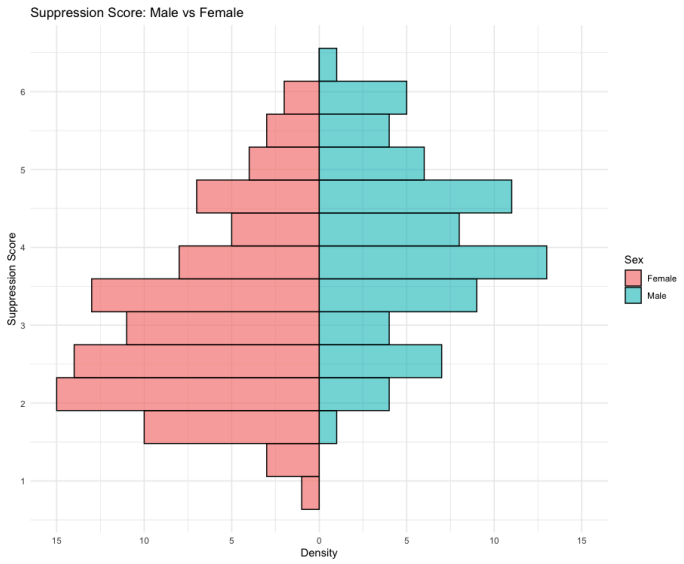
As mentioned earlier, this paper will focus on the mean of the participant's suppression score ($M = 3.53$, $SD = 1.21$) and the participant's sex. Before carrying out the *t*-test, we will need to determine if the two sex groups; male ($M = 4.03$, $SD = 1.08$) and female ($M = 3.15$, $SD = 1.17$) are normally distributed. After applying the Sharpiro-Wilk test of normality, the male group ($p = 0.58$) had a p-value greater than 0.05, indicating a normal distribution. In comparison, the female group ($p = 0.02$) had a p-value less than 0.05, indicating a distribution that is not normal. This is also visually displayed in Figure 2a and Figure 2b with the Q-Q plot for both sexes. However, because each sex group has a large sample size (73 males, 97 females), we can still conduct the *t*-test.

Additionally, since the standard deviation for males ($SD = 1.08$) and females ($SD = 1.17$) are different, this further indicates that we can use Welch's *t*-test.

Results

Figure 1

Comparative Density Histogram of Self-Reported Suppression Score by Sex



Note. Single outlier observed in female group

For our null hypothesis, we assume there will be no difference between the groups. Our alternative hypothesis assumes there is a difference between the male group and the female group which is greater than 0.

In the MIDUS 3 dataset, male participants were found to have a higher suppression score ($M = 4.03$, $SD = 1.08$) than female participants ($M = 3.15$, $SD = 1.17$), $t(167) = 5.03$, $p < .001$, 95% confidence interval [0.59, ∞]. From this data, we have evidence to reject the null hypothesis.

Discussion

From these findings, male participants had a higher suppression score than female participants which aligned with previous literature. This supports the idea that men are more prone to suppress their emotions due to biological and social factors that make it difficult to develop healthy emotional regulation techniques.

While our findings do support our hypothesis, there are possible limitations that are best to consider for future research. Firstly, MIDUS 3's ERQ was a self-reported dataset which is exposed to potential response bias. Male participants may overestimate how much and how well they suppress their emotions and give themselves a higher suppression score. While female participants may have underestimated their suppression score. It is recommended that future studies should implement a way to restrict this potential bias with additional measures or approach a different method to record suppression scores that are not reliant on participant perspective.

Secondly, there may be confounding variables that may have impacted the results. For instance, age can influence how one may regulate their emotions. As the younger generation challenges the concept of masculinity, they are more likely to have a lower mean score in contrast to the older generation, who were raised on this masculine mindset that a man cannot show any emotions. To minimize the influence of this confounding variable, future research should consider using a large population with a diverse age range.

References

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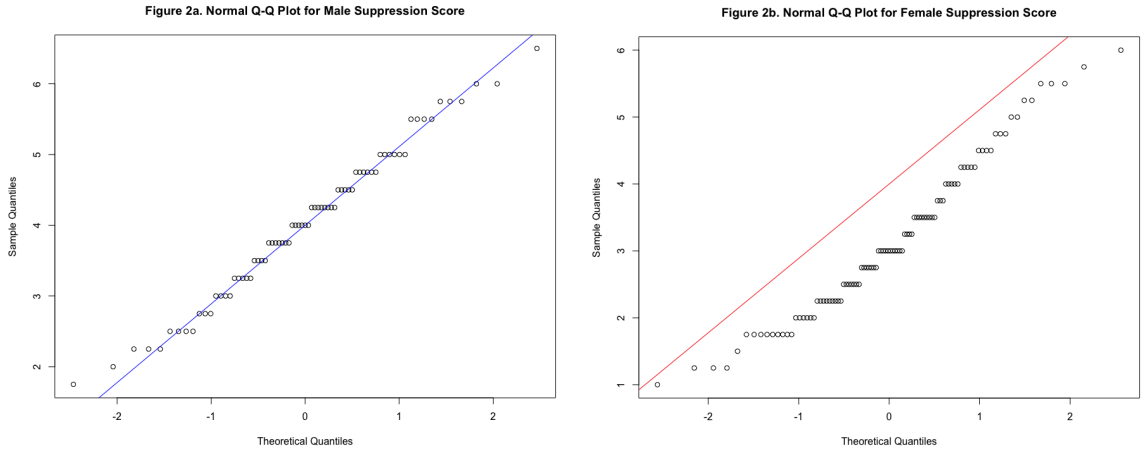
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Appendix

Figure 2

Q-Q Plots for Male Suppression Score and Female Suppression Score



Note. Figure 2a represents the male participant group (n = 73). The data points closely follow the line of normality, indicating a normal distribution. Figure 2b represents the female participant group (n = 97). The data points deviate below the line, indicating a non-normal distribution.